

Witty Machines AG
Zeughausstrasse 31
8004 Zürich
Switzerland

May 2, 2025

To Whom It May Concern,

I am pleased to recommend Mr. Julien Stocher, who completed a robotics software engineering internship at Witty Machines from February 1, 2025, to May 2, 2025. Witty Machines is a Swiss robotics startup specializing in case-packing robots for the food industry, and during his time with us, Julien made valuable contributions to our software team.

Scope of Work and Key Achievements

- **RFID Integration:** Julien led the integration of an RFID reader to track Reusable Packaging Units. He not only ensured data capture from the hardware sensor but also architected the software interfaces that allow our robots to register and log each packaging unit.
- **Field Deployment Operations:** He helped operate our pilot robot during a live deployment with one of our food-industry partners. He not only ensured that the robot was operating smoothly, but also learned how to troubleshoot other machines on site to increase the overall throughput. Furthermore, he carefully observed the manual packing process performed by the workers and, based on that, suggested improvements for the robot process that led to a >200% increase in packing speed.
- **Computer Vision Model for Label Inspection:** Julien trained and validated a convolutional neural network to detect printed versus mis-printed labels on egg cartons. He then integrated this model into our ROS-based control stack, enabling real-time quality inspection during packing.
- **Instance Segmentation and Localization for Packaging Units:** Julien collected and annotated data on boxes from different vendors and trained models to detect them in images and localize them. Over multiple iterations, the models were trained and were ultimately deployed in the production system.

Professional Traits and Team Fit

Julien consistently impressed us with his diligent work ethic, technical versatility, and eagerness to learn. He approached each challenge with a positive attitude, often volunteering to stay late to refine algorithms or to deepen his expertise in a particular subject.

Conclusion and Recommendation

During his internship, Julien established a solid grounding in robotics-software and proved eager to keep learning. His practical experience, collaborative spirit, and proactive attitude would benefit any early-career role in robotics or machine-learning development. I recommend him with confidence and am happy to provide further details on request.

Sincerely,



Johannes Pankert

CTO

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Harvard
School of Engineering and
Applied Sciences

Katia Bertoldi
Professor of Applied Mechanics
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September 5, 2024

To Whom It May Concern,

It is a pleasure to be able to write in support of Christopher Julien Stocker Salas. By way of introduction, I am the William and Ami Kuan Danoff Professor of Applied Mechanics at the School of Engineering and Applied Sciences at Harvard University. In my research, I combine theoretical, computational and experimental methods to gain deeper insight into the non-linear behavior of materials and structures. Guided by theoretical and numerical analysis, we exploit material and geometric non-linearities to design novel materials with tunable exceptional properties, and we seek to create and test such materials.

Christopher Julien Stocker Salas conducted his Master's thesis research under my supervision at the Materials and Structures by Design Laboratory, Harvard University, from June 6 to December 12, 2023. During this period, Christopher's research focused on "Analog Fluidic Networks: Towards Computational Non-Electronic Systems," an innovative and challenging project that explored alternative methods for computational systems in the field of soft robotics.

Christopher's thesis aimed to develop and characterize fluidic networks capable of performing mechanical computations using analog logic. His approach involved leveraging fluidic components that could execute both linear and nonlinear functions. The project required Christopher to combine theoretical insights with practical experimentation, using various sensors and custom hardware setups to validate our mathematical models. Throughout his thesis work, Christopher displayed exceptional dedication, creativity, and problem-solving skills. He was responsible for the comprehensive characterization of the fluidic network's elements, developing methods to measure and analyze the flow vs pressure responses of these components at various points of the network. His experimental work provided insights into how these analog networks could be harnessed to perform entirely mechanical calculations, demonstrating their potential to offload rigid electronic control units to soft mechanical control in the field of soft robotics.

This research is significant for the development of soft robotic systems, particularly in applications where flexibility and compliance are paramount. His ability to bridge concepts from materials science,

information processing, and robotics highlights his interdisciplinary approach and deep understanding of the subject matter.

In addition to his technical abilities, Christopher's multicultural background allows him to integrate and collaborate well across languages. He consistently demonstrated professionalism, initiative, and a collaborative spirit, which greatly contributed to the advancement of his research. His strong analytical skills, coupled with his ability to hands-on approach, make him valuable member of any team. I have no doubt that Christopher will excel in his future endeavors and make significant contributions to his team. I highly recommend him for any opportunities that align with his skills and interests.

Best regards

Katia Bertoldi

Katia Bertoldi

William and Ami Kuan Danoff Professor of Applied Mechanics

School of Engineering and Applied Sciences, Harvard University

**To the Members of
Bridge - Proof of Concept
Evaluation Panel**

Reconfigurable Robotics Lab
EPFL
Lausanne
Switzerland

September 5th, 2024

To Whom It May Concern,

I am writing to provide a reference for Christopher Julien Stocker Salas, who conducted his Master's thesis research and Master's semester project under my supervision at the Reconfigurable Robotics Lab at EPFL. His Master's thesis was done at Harvard University in collaboration with Katia Bertoldi's group from June 6 to December 12, 2023. During this period, Christopher's research focused on "Analog Fluidic Networks: Towards Computational Non-Electronic Systems," an innovative and challenging project that explored alternative methods for computational systems in the field of soft robotics.

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Best regards,



Jamie Paik, PhD
Head of the Reconfigurable Robotics Lab